

- 5** **Table 1** shows the taxons and the names of the taxons used to classify one species of otter. They are **not** in the correct order.

Table 1

	Taxon	Name of taxon
J	Family	Mustelidae
K	Kingdom	Animalia
L	Genus	Lutra
M	Class	Mammalia
N	Order	Carnivora
O	Phylum	Chordata
P	Domain	Eukarya
Q	Species	lutra

- 0 5** . **1** Put letters from **Table 1** into the boxes in the correct order. Some boxes have been completed for you.

[1 mark]

		O	M			L	Q
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- 0 5** . **2** Give the scientific name of this otter.

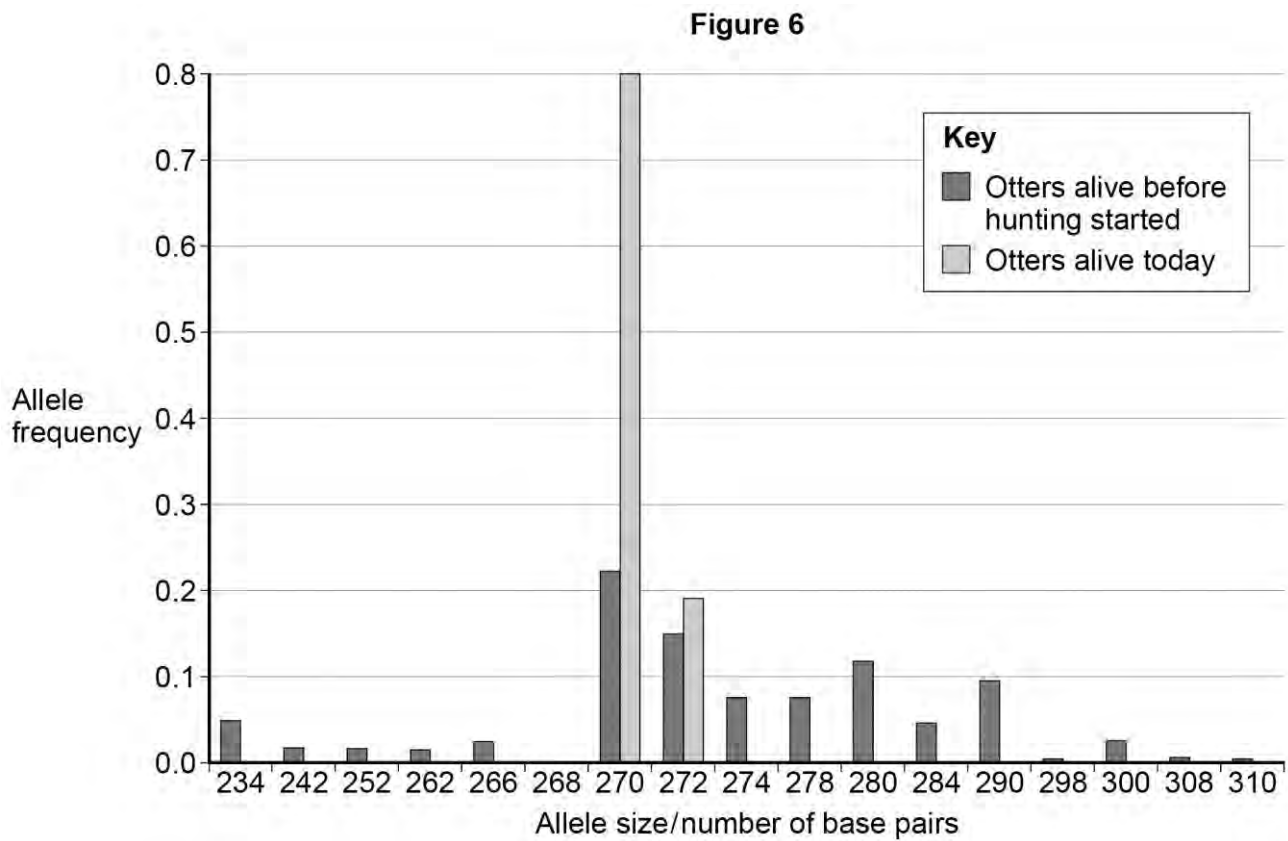
[1 mark]

Scientists investigated the effect of hunting on the genetic diversity of otters. Otters are animals that were killed in very large numbers for their fur in the past.

The scientists obtained DNA from otters alive today and otters that were alive before hunting started.

For each sample of DNA, they recorded the number of base pairs in alleles of the same gene. Mutations change the numbers of base pairs over time.

Figure 6 shows the scientists' results.



0 5 . 3 The scientists obtained DNA from otters that were alive before hunting started.

Suggest **one** source of this DNA.

[1 mark]

0 5 . 4 What can you conclude about the effect of hunting on genetic diversity in otters? Use data from **Figure 6** to support your answer.

[2 marks]

Question 5 continues on the next page

Turn over ►

0 5 . 5 Some populations of animals that have never been hunted show very low levels of genetic diversity.

Other than hunting, suggest **two** reasons why populations might show very low levels of genetic diversity.

[2 marks]

1 _____

2 _____
